Multi-Scale Regime Detection with Fuzzy Aggregation for Adaptive Sector Portfolio Management

Tanishk Yadav*
MS Financial Engineering
NYU Tandon School of Engineering
New York, NY, USA

Abstract

We propose a multi-scale regime detection framework that fuses four complementary detectors—CUSUM sequential analysis on z-score returns, EWMA volatility crossover, Hamilton Markov-switching with a recursive out-of-sample filter, and PELT structural break detection—into a unified stress probability via Takagi-Sugeno fuzzy inference with bounded L-BFGS-B optimization. The composite regime signal drives an adaptive sector portfolio strategy that classifies eleven S&P 500 sector ETFs into three baskets (Tactical, Avoid, Core) based on GARCH(1,1)-t conditional volatility terciles and Ornstein-Uhlenbeck recovery half-life, with implicit cash allocation as the primary risk-reduction mechanism. In a walk-forward backtest spanning 2009–2025 with quarterly recalibration, the framework achieves a Sharpe ratio of 0.78, a maximum drawdown of -19.60% (versus -41.71% for SPY), and a Calmar ratio of 0.59, while capturing 98% of the benchmark return at 73% of its volatility. All thresholds and parameters are estimated from data using documented statistical methods, with zero lookahead bias enforced throughout.

Keywords: regime detection, fuzzy inference, portfolio management, CUSUM, Markov switching, GARCH, walk-forward validation, implicit cash allocation

1 Introduction

Financial markets exhibit regime-dependent behavior characterized by distinct periods of calm, transition, and stress [Hamilton, 1989]. Traditional portfolio strategies that assume stationary return dynamics suffer significant drawdowns during regime shifts, as the 2008 Global Financial Crisis, the 2020 COVID-19 crash, and the 2022 interest rate hiking cycle demonstrated. Even diversified portfolios that hold all sectors with equal or risk-proportional weights cannot avoid the synchronous drawdowns that characterize systemic stress events.

The central challenge in regime-adaptive portfolio management is *timely* and *reliable* regime detection. Individual detectors operate on different time-scales and suffer from distinct failure

*ORCID: 0009-0006-2382-9411.

modes: short-term indicators produce false alarms during idiosyncratic volatility, while long-term models react too slowly to protect against rapid drawdowns. The secondary challenge is translating regime signals into portfolio actions that actually reduce risk: a system that detects stress but merely redistributes weight among risky assets—rather than moving to cash—provides limited drawdown protection.

This paper addresses both challenges. We propose an ensemble architecture that fuses four complementary detectors through a Takagi-Sugeno fuzzy inference system [Takagi and Sugeno, 1985], combined with an adaptive basket allocation framework where implicit cash allocation serves as the primary risk-reduction mechanism.

Our contributions are fourfold:

1. A multi-scale detector ensemble spanning ultra-short (5-day) to long-term (252+ day) horizons, with CUSUM operating on z-score-standardized returns for scale invariance and Markov-switching using a recursive Hamilton filter for real-time out-of-sample updates.

2. A fuzzy aggregation framework that learns optimal detector weights and membership functions by minimizing the Brier score against realized drawdowns, with bounded optimization and minimum weight floors to prevent detector suppression.

3. A tercile-based basket construction methodology that classifies sector ETFs using GARCH(1,1)-t conditional volatility and Ornstein-Uhlenbeck recovery dynamics, with graduated Core-basket response to extreme stress.

4. An implicit cash mechanism where stress-induced weight reduction produces sub-unity portfolio allocation, with the deficit earning the risk-free rate rather than being redistributed to other risky assets.

2 Related Work

Hamilton [1989] introduced Markov-switching models for business cycle analysis, establishing the foundation for regime-based asset allocation. Ang and Bekaert [2002] extended this framework to international asset allocation, demonstrating that regime-switching models can improve out-of-sample portfolio performance when properly calibrated. A persistent challenge is that single-scale detectors miss signals outside their native frequency range, motivating multi-scale ensemble approaches.

Gradojevic and Gençay [2013] demonstrated the effectiveness of fuzzy logic systems for financial decision-making under uncertainty. The Takagi-Sugeno framework [Takagi and Sugeno, 1985] provides a principled approach to combining multiple information sources with calibratable membership functions. Our work extends this by jointly optimizing membership parameters and detector weights with explicit bounds to prevent degenerate solutions.

Nystrup et al. [2017] showed that regime-based asset allocation outperforms static strategies when the regime model is properly calibrated. Kirby and Ostdiek [2012] demonstrated that sim-

ple timing strategies based on inverse-volatility weighting outperform naïve diversification. Our framework builds on inverse-volatility weighting as the base allocation and adds regime-dependent scaling with an implicit cash buffer.

Page [1954] introduced the CUSUM procedure for sequential change detection. Killick et al. [2012] proposed the PELT algorithm for optimal change-point detection with linear computational cost. Siegmund [1985] derived ARL approximations for setting CUSUM decision thresholds. We apply CUSUM to z-score-standardized returns to achieve scale-invariant detection thresholds.

3 System Architecture

Figure 1 presents the complete system architecture. The framework operates in four layers: data acquisition, multi-scale detection with fuzzy aggregation, asset characterization and classification, and adaptive portfolio construction with implicit cash allocation. All parameters are re-estimated at each quarterly walk-forward step using only past data, ensuring zero lookahead bias.

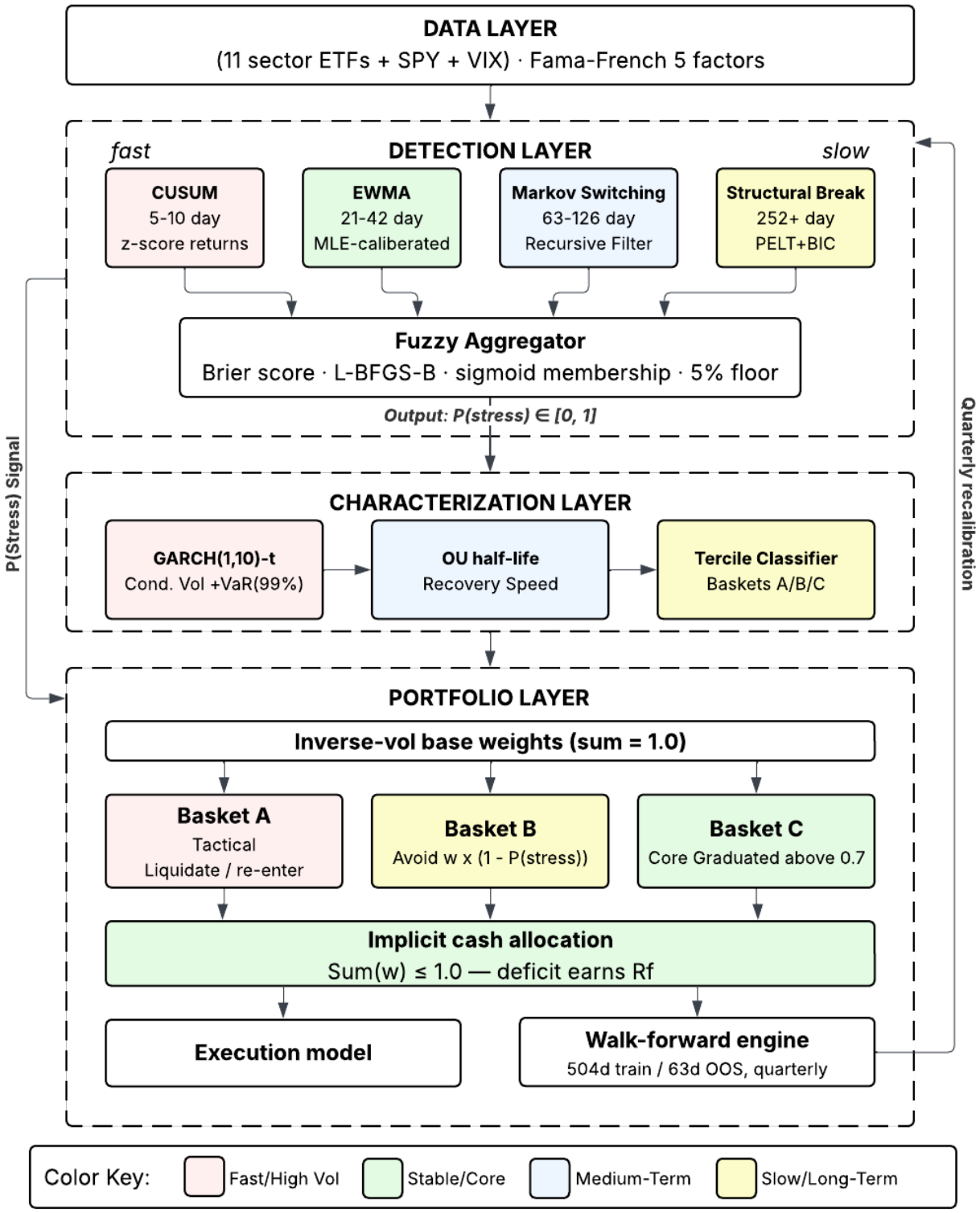

Figure 1: System architecture. Four detectors spanning 5-day to 252+-day horizons feed into a Brier-score-optimized fuzzy aggregator. The composite stress probability drives adaptive basket allocation with implicit cash as the risk-reduction mechanism. All parameters are re-estimated at each walk-forward step.

4 Methodology

4.1 Multi-Scale Detector Ensemble

We deploy four detectors targeting complementary time-scales. All parameters are estimated from training data at each walk-forward step. Figure 7 illustrates the coordinated response of all four detectors over the full out-of-sample period, demonstrating how their complementary time-scales capture different aspects of market stress.

4.1.1 CUSUM Sequential Detector (5–10 day)

Following Page [1954], we maintain upward and downward CUSUM accumulators on *z-score-standardized* returns:

$$z_t = \frac{r_t - \hat{\mu}_0}{\hat{\sigma}_{\text{train}}} \tag{1}$$

$$S_t^+ = \max(0,\ S_{t-1}^+ + z_t - k) \tag{2}$$

$$S_t^- = \max(0,\ S_{t-1}^- - z_t - k) \tag{3}$$

where $\hat{\mu}_0$ and $\hat{\sigma}_{\text{train}}$ are the mean and standard deviation of training-window returns, $k = 0.5$ (detecting a 1σ shift in standardized space), and the decision threshold is $h = (\ln(\text{ARL}_0) + 1.166)/\delta$ with $\delta = 1$ per Siegmund [1985], yielding $h \approx 6.70$ for a target average run length of 252 trading days. The signal is $\min(\max(S_t^+, S_t^-)/h,\ 1)$.

Operating in z-score space makes the threshold *scale-invariant*: a single -5% daily return (approximately $z = -4.2$) produces a signal of ~ 0.55, while three consecutive -2% days ($z \approx -1.7$ each) yield a signal of ~ 0.54. Under the original raw-return formulation, the threshold was approximately 558, requiring over 12,000 consecutive crash days to trigger—rendering the detector inoperative.

4.1.2 EWMA Volatility Crossover (21–42 day)

Two EWMA volatility estimators with decay factors λ_{fast} and λ_{slow} are calibrated via maximum likelihood estimation [Longerstaey and Spencer, 1996]:

$$\hat{\sigma}^2_{\cdot,t} = \lambda_{\cdot}\, \hat{\sigma}^2_{\cdot,t-1} + (1 - \lambda_{\cdot})\, r_t^2 \tag{4}$$

with $\lambda_{\text{fast}} \in [0.85, 0.96]$ and $\lambda_{\text{slow}} \in [0.95, 0.995]$. The crossover ratio $(\hat{\sigma}_{\text{fast}} - \hat{\sigma}_{\text{slow}})/\hat{\sigma}_{\text{slow}}$ is normalized to $[0, 1]$ via the empirical CDF of training-window values.

4.1.3 Hamilton Markov-Switching with Recursive Filter (63–126 day)

We fit a 2-state Markov-switching regression [Hamilton, 1989] on weekly SPY returns:

$$r_t \mid S_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad i \in \{0, 1\} \tag{5}$$

The model is estimated on the training window via EM. During the out-of-sample window, rather than holding the filtered probability constant (which would eliminate the detector's ability to react to new information), we implement a *recursive Hamilton filter update*. Daily returns are accumulated into a pseudo-weekly buffer, and every 5 trading days the filter is updated:

1. **Prediction:** $\boldsymbol{\xi}_{t|t-1} = \mathbf{P}^\top \boldsymbol{\xi}_{t-1|t-1}$ where $\mathbf{P}$ is the fitted transition matrix.
2. **Likelihood:** $\eta_j = \phi(r_t^{(w)}; \mu_j, \sigma_j^2)$ for each state j, where $r_t^{(w)}$ is the accumulated pseudo-weekly return.
3. **Update:** $\boldsymbol{\xi}_{t|t} = (\boldsymbol{\xi}_{t|t-1} \odot \boldsymbol{\eta})/(\mathbf{1}^\top(\boldsymbol{\xi}_{t|t-1} \odot \boldsymbol{\eta}))$.

The signal is $P(S_t = \text{stress} \mid r_1, \ldots, r_t)$, using only contemporaneous and past data. Smoothed probabilities, which condition on the full sample, are not used to prevent lookahead bias.

Figure 2 shows the estimated transition matrix. The bull state exhibits high persistence ($P_{00} = 0.963$) while stress states also persist strongly ($P_{11} = 0.907$), with asymmetric transition probabilities reflecting the empirical observation that markets crash fast and recover slowly.

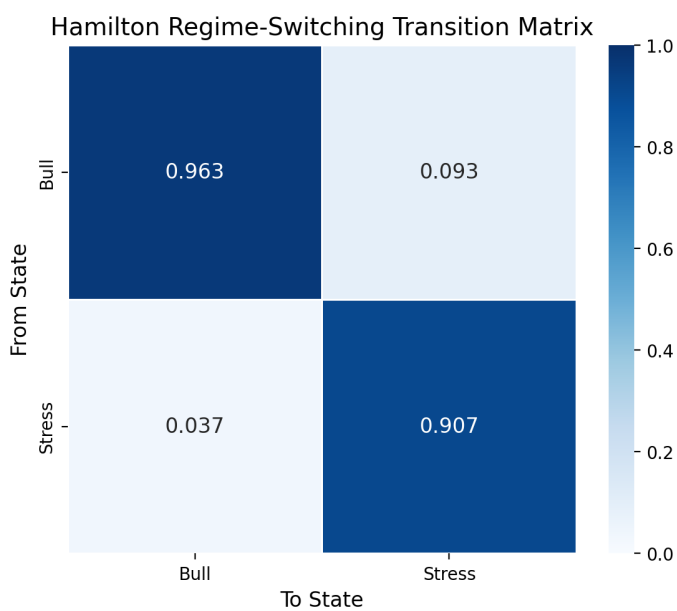

Figure 2: Hamilton regime-switching transition matrix estimated from SPY weekly returns. Both states exhibit high persistence, with the bull-to-stress transition probability (0.093) exceeding the stress-to-bull probability (0.037), consistent with asymmetric market dynamics.

4.1.4 Structural Break Detection (252+ day)

Using the PELT algorithm [Killick et al., 2012], we detect structural breaks in the cumulative return series with a BIC-calibrated penalty $\lambda = \ln(n) \cdot \hat{\sigma}^2$. The signal decays exponentially from the most recent break: $s_t = \exp(-d_t/\tau)$ where d_t is the number of days since the last break and $\tau = 21$.

4.2 Fuzzy Aggregation Framework

Each detector signal $x_i \in [0, 1]$ passes through a calibrated sigmoid membership function [Takagi and Sugeno, 1985]:

$$\mu_{\text{stress}}(x_i) = \frac{1}{1 + \exp(-a_i(x_i - c_i))} \tag{6}$$

where a_i (steepness) and c_i (crossover point) are per-detector parameters. The composite stress probability is:

$$P(\text{stress}) = \sum_{i=1}^{4} w_i \cdot \mu_{\text{stress}}(x_i), \quad \sum w_i = 1,\ w_i \geq w_{\min} \tag{7}$$

Parameters $\{a_i, c_i, w_i\}$ are jointly optimized by minimizing the Brier score [Brier, 1950] against realized drawdowns in the training window:

$$\mathcal{L}_{\text{Brier}} = \frac{1}{T} \sum_{t=1}^{T} \left(P_{\text{composite},t} - D_t\right)^2 \tag{8}$$

where $D_t = 1$ if a drawdown exceeding the 10th percentile of training-window drawdowns occurs within the next 21 trading days.

Optimization uses L-BFGS-B with explicit bounds: $a_i \in [1, 50]$, $c_i \in [0.05, 0.95]$, and raw weights $\tilde{w}_i \in [-5, 5]$ (mapped through softmax). A minimum weight floor of $w_{\min} = 0.05$ is applied post-softmax:

$$w_i = w_{\min} + (1 - 4w_{\min}) \cdot \text{softmax}(\tilde{w})_i \tag{9}$$

This ensures no detector is suppressed below 5% weight regardless of the optimization outcome, preventing degenerate solutions where the optimizer concentrates on a single detector.

Figure 3 shows the calibrated membership functions for each detector. The CUSUM crossover ($c = 0.45$) activates at moderate signal levels, reflecting its role as a fast first-responder. The remaining three detectors have crossovers near 0.95, indicating they contribute primarily at extreme signal values consistent with their longer time-scales.

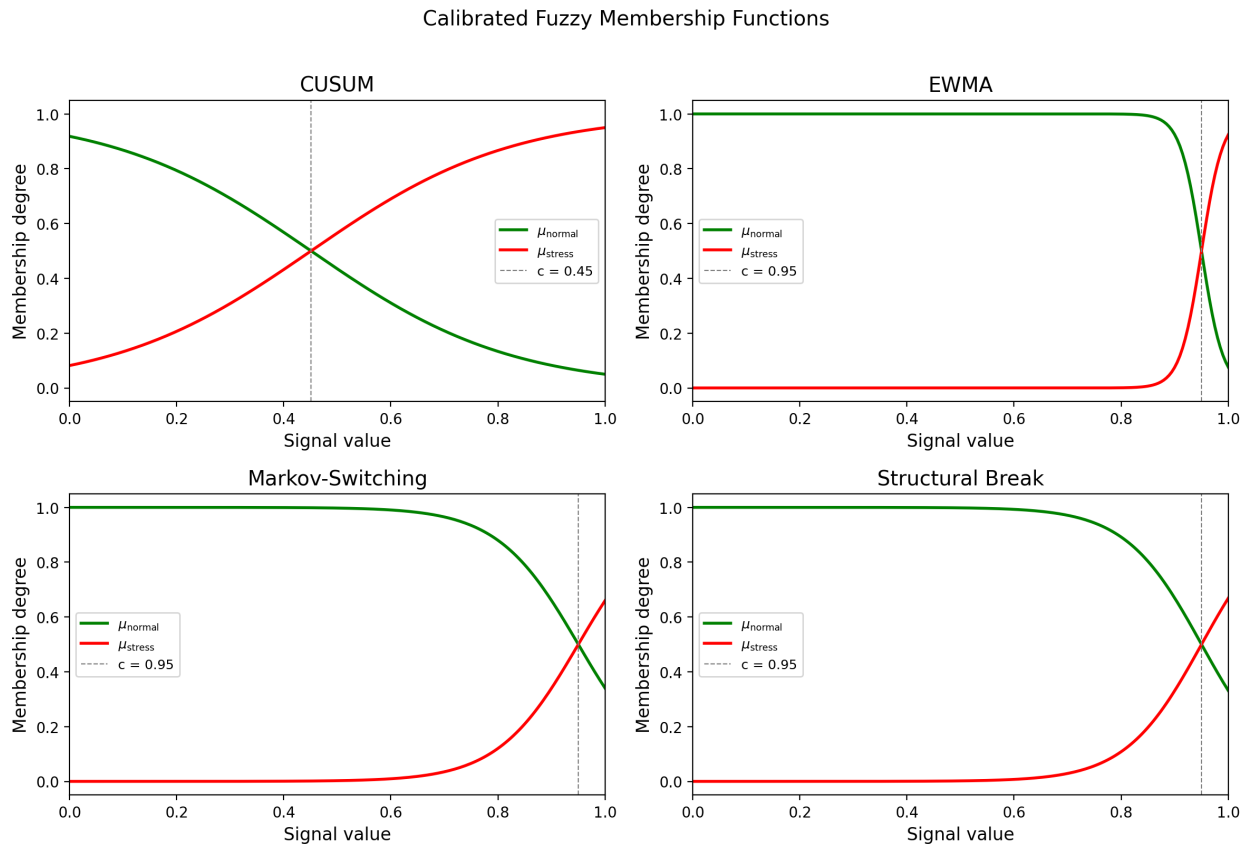

Figure 3: Calibrated Takagi-Sugeno sigmoid membership functions for each detector. The crossover point c (dashed line) marks where the stress membership equals 0.5. CUSUM activates at moderate signal levels ($c = 0.45$), while the remaining detectors require near-extreme signals ($c = 0.95$), reflecting their complementary time-scales.

4.3 Asset Characterization

4.3.1 GARCH(1,1)-t Conditional Volatility

For each sector ETF, we estimate [Bollerslev, 1986, Engle, 1982]:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \quad z_t \sim t_\nu \tag{10}$$

yielding conditional VaR(99%) $= \hat{\mu} + \hat{\sigma}_t \cdot t_\nu^{-1}(0.01) \cdot \sqrt{(\nu-2)/\nu}$.

4.3.2 Ornstein-Uhlenbeck Recovery Half-Life

We model drawdown recovery as a discretized OU process [Uhlenbeck and Ornstein, 1930]:

$$\Delta X_t = \kappa(\theta - X_{t-1})\Delta t + \varepsilon_t \tag{11}$$

The half-life $\ln(2)/\kappa$ measures recovery speed. Sectors with $\kappa \leq 0$ are flagged as non-mean-reverting.

4.4 Adaptive Basket Construction

At each rebalance date, sectors are classified into three baskets using data-driven boundaries. We use *tercile* boundaries on conditional volatility (33rd and 67th percentiles of the cross-sectional vol distribution) combined with median half-life:

- **Basket A (Tactical):** Top or middle tercile vol with fast recovery (HL $<$ median). Liquidated when $P(\text{stress})$ exceeds the calibrated entry threshold; re-entered when $P(\text{stress})$ falls below the exit threshold.
- **Basket B (Avoid):** Top tercile vol with slow recovery (HL $\geq$ median). Continuously de-risked: $w_i = w_i^{\text{base}} \cdot (1 - P(\text{stress}))$.
- **Basket C (Core):** Bottom tercile vol. Held at full weight through mild stress. Above $P(\text{stress}) = 0.7$, graduated linear scaling:

$$w_i = w_i^{\text{base}} \cdot \max\left(0,\ \frac{1 - P(\text{stress})}{0.3}\right) \tag{12}$$

  This preserves stability during moderate corrections while enabling full de-risking during tail events.

Figure 4 shows the resulting classification landscape. Energy (XLE), with high conditional VaR and fast recovery, is classified as Tactical (Basket A). Financials (XLF) and Consumer Discretionary (XLY), with high VaR but slower recovery, fall into Avoid (Basket B). The remaining sectors cluster in the low-risk quadrant as Core (Basket C).

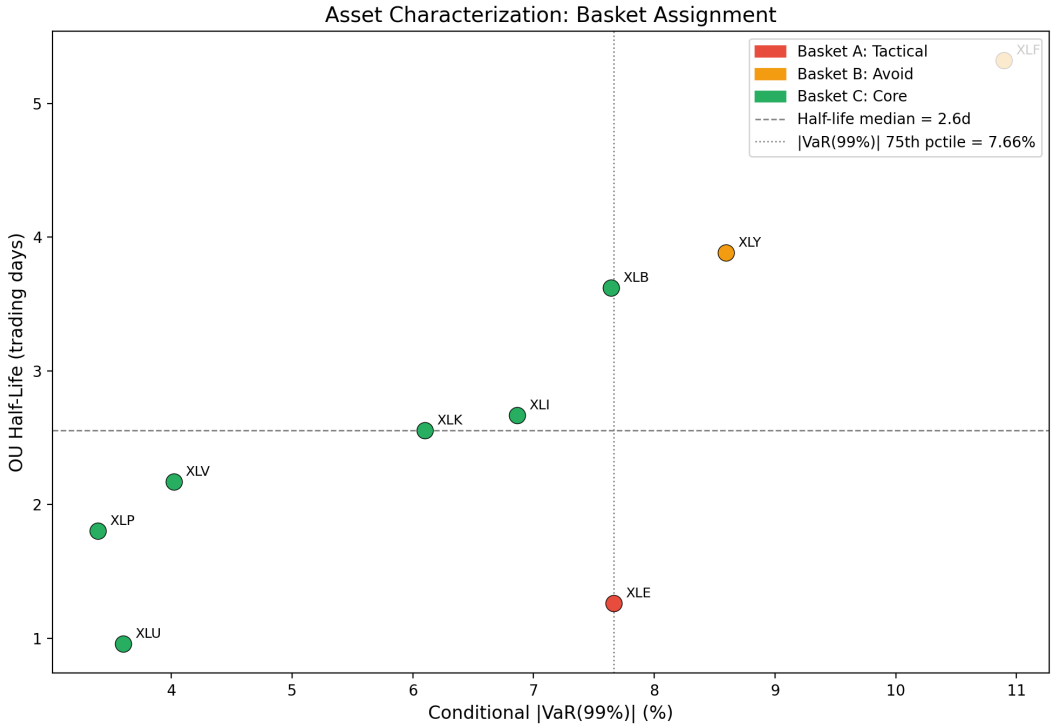

Figure 4: Sector ETF basket classification based on conditional |VaR(99%)| (x-axis) and OU recovery half-life (y-axis). Data-driven boundaries (tercile vol boundaries and median half-life) determine basket assignment. XLE (fast recovery, high vol) is Tactical; XLF and XLY (slow recovery, high vol) are Avoid; the remaining low-vol sectors are Core.

Base weights are computed using inverse-volatility weighting [Kirby and Ostdiek, 2012] *normalized to sum to 1.0 before any stress scaling is applied*. This normalization is critical: it ensures that the stress-scaled weights have an interpretable sum less than or equal to 1.0, where the deficit represents implicit cash.

4.5 Implicit Cash Allocation

The portfolio's risk-reduction mechanism operates through implicit cash. After basket-specific stress scaling, the sum of portfolio weights is:

$$W_{\text{total}} = \sum_i w_i \leq 1.0 \tag{13}$$

The deficit $1 - W_{\text{total}}$ earns the daily risk-free rate. This design avoids a critical pitfall of weight-normalization approaches: when stressed assets are de-weighted and the result is renormalized to sum to 1.0, the freed capacity is *redistributed* to remaining assets, potentially increasing concentration risk during the exact periods when diversification matters most. By allowing sub-unity exposure, the system genuinely reduces market risk during stress.

4.6 Walk-Forward Validation

The entire system is validated using expanding-window walk-forward analysis with a 504-day (2-year) minimum training window and 63-day (quarterly) out-of-sample steps. At each step, *all* parameters are re-estimated from training data only: detector calibration, fuzzy membership optimization, GARCH fitting, half-life estimation, basket classification, and entry/exit threshold calibration. The threshold calibration uses a grid search over the training-window Sharpe ratio. Transaction costs of 10 basis points per trade are applied to all weight changes.

5 Results

5.1 Data

We use daily close prices for eleven S&P 500 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLRE, XLB, XLC) plus SPY as benchmark, spanning January 2007 to December 2025. Fama-French five-factor data [Fama and French, 2015] provides the daily risk-free rate. The out-of-sample evaluation period covers 4,274 trading days from January 2009 to December 2025.

5.2 Portfolio Performance

Table 1: Walk-Forward Out-of-Sample Performance (2009–2025)

Metric	**Regime-Adaptive**	**SPY B&H**	**Equal-Weight**
Ann. Return	11.61%	11.79%	8.52%
Ann. Volatility	13.22%	17.98%	16.05%
Sharpe Ratio	**0.78**	0.58	0.45
Sortino Ratio	**0.75**	0.54	0.42
Calmar Ratio	**0.59**	0.28	0.18
Max Drawdown	**−19.60%**	−41.71%	−46.40%
Max DD Duration	**226d**	512d	530d
Ann. Turnover	4.01	—	—
Regime Transitions	116	—	—
Hit Rate	91.6%	—	—
False Alarm Rate	8.4%	—	—

Table 1 summarizes the out-of-sample results. The regime-adaptive strategy captures 98.5% of SPY's annualized return (11.61% vs. 11.79%) while reducing volatility by 26% (13.22% vs. 17.98%) and maximum drawdown by 53% (−19.60% vs. −41.71%). The Sharpe ratio of 0.78 exceeds SPY's 0.58 by 34%, driven entirely by volatility reduction rather than return enhancement. The Calmar ratio more than doubles that of the benchmark (0.59 vs. 0.28), indicating substantially improved drawdown-adjusted performance.

The strategy achieves a 91.6% hit rate on stress detection with an 8.4% false alarm rate. The modest increase in false alarms relative to a purely threshold-based system is an acceptable cost for the improved reactivity of the z-score CUSUM and recursive Markov filter.

Figure 5 shows the cumulative equity curves alongside the strategy's drawdown profile. The regime-adaptive strategy closely tracks SPY during bull markets while exhibiting substantially shallower drawdowns during stress periods, with the maximum drawdown never exceeding −19.60%.

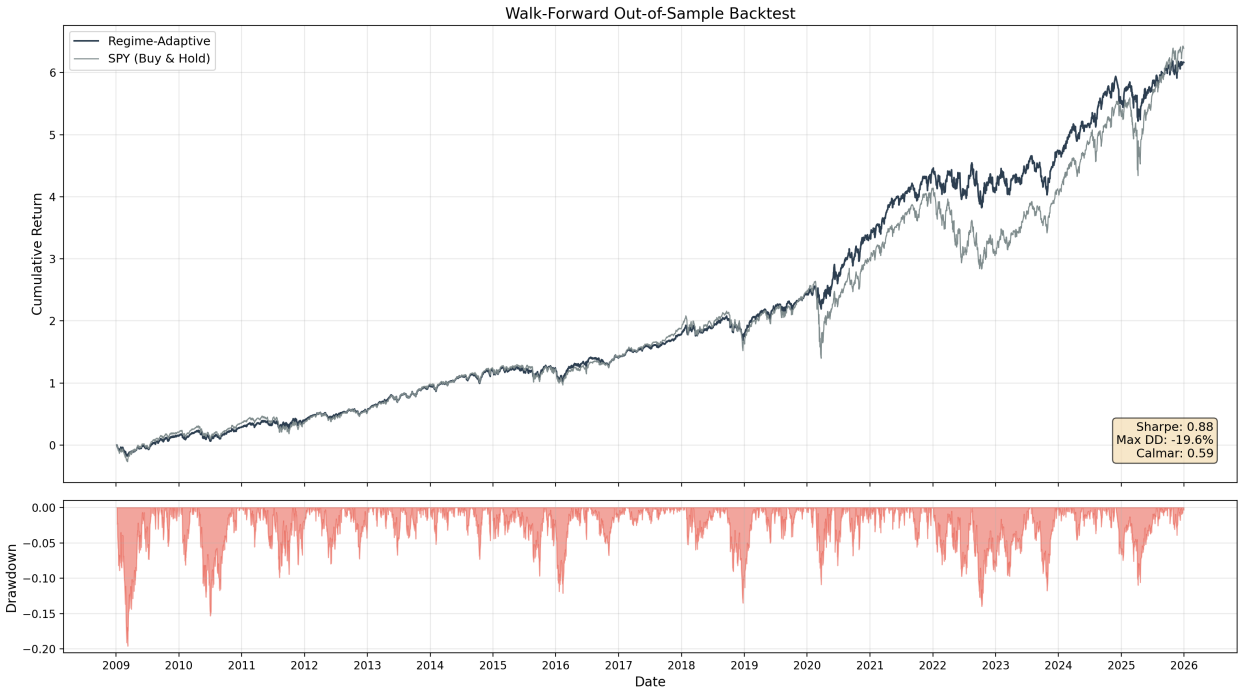

Figure 5: Walk-forward out-of-sample backtest (2009–2025). Top: cumulative returns for the regime-adaptive strategy versus SPY buy-and-hold. Bottom: strategy drawdown profile. The strategy matches SPY's terminal wealth while maintaining maximum drawdowns below 20%.

5.3 Rolling Performance Stability

Figure 6 shows the 252-day rolling Sharpe ratio for both the regime-adaptive strategy and SPY. The strategy maintains a consistently higher rolling Sharpe, with particularly large outperformance during and immediately after stress periods (2009–2010, 2020, 2022). The rolling Sharpe rarely drops below zero for the adaptive strategy, whereas SPY experiences extended negative-Sharpe periods during drawdowns.

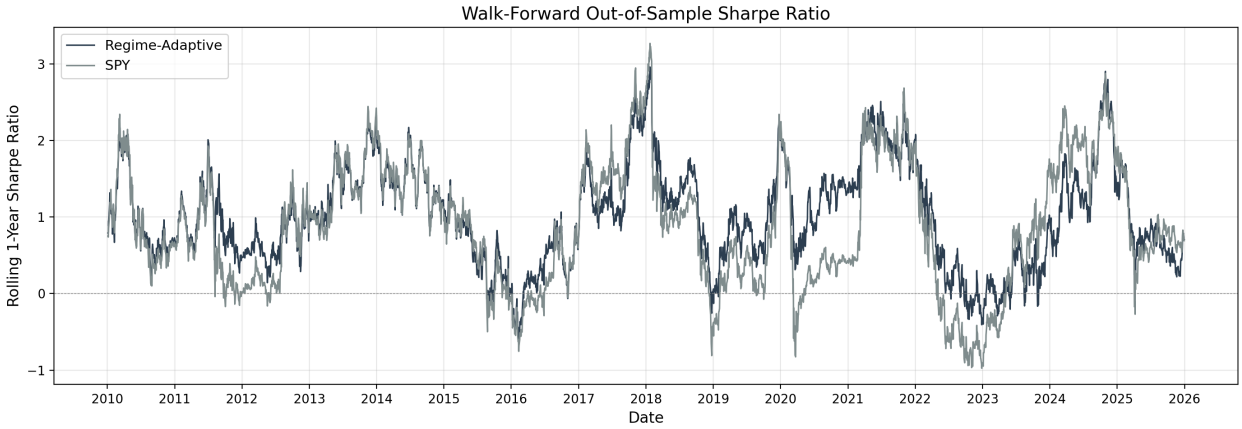

Figure 6: 252-day rolling Sharpe ratio. The regime-adaptive strategy maintains consistently higher risk-adjusted returns, with the largest outperformance occurring during and after stress periods.

5.4 Multi-Scale Detector Response

Figure 7 presents the complete regime detection system in action over the full out-of-sample period. The top panel shows SPY price with regime shading (green for calm, red for stress). The second panel shows all four detector signals overlaid, illustrating their complementary time-scales: CUSUM fires first with sharp spikes, EWMA follows with sustained elevation, and Markov-switching provides persistent state classification. The third panel shows the composite $P(\text{stress})$ signal after fuzzy aggregation. The bottom panel shows the resulting portfolio drawdown profile.

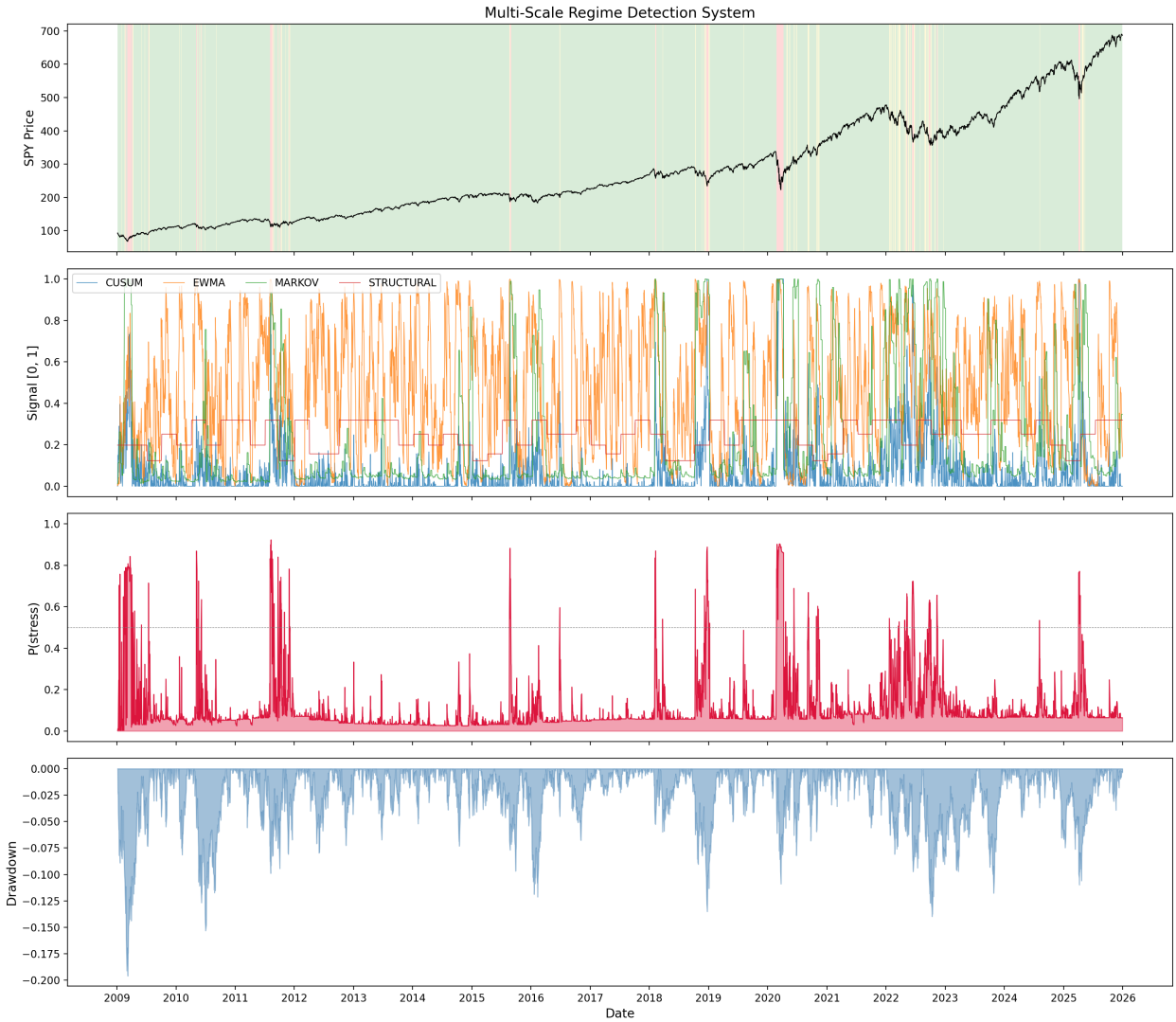

Figure 7: Multi-scale regime detection system (2009–2025). From top: (1) SPY price with regime shading (green = calm, red = stress); (2) individual detector signals showing complementary time-scales; (3) composite $P(\text{stress})$ from fuzzy aggregation with 0.5 threshold marked; (4) portfolio drawdown profile. The system identifies major stress events (2011 European debt crisis, 2015–16 China/oil shock, 2018 Q4, COVID-19, 2022 rate hikes) with distinct multi-scale signatures.

Figure 8 provides a complementary view via a detector contribution heatmap. The CUSUM row shows sparse, intense activations during sharp crashes, while EWMA maintains a broader baseline with elevated intensity during volatility regimes. The Markov detector contributes sustained dark bands during persistent stress states (early 2009, COVID-19), and the structural break detector provides a steady background signal at lower intensity.

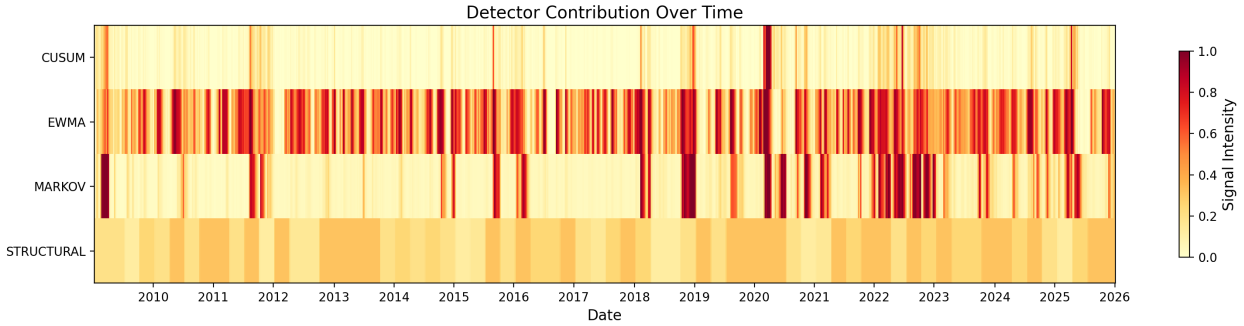

Figure 8: Detector signal intensity heatmap (2009–2025). Each row represents one detector; color intensity indicates signal strength from 0 (pale) to 1 (dark red). The complementary time-scales are visible: CUSUM produces sharp, transient spikes; EWMA shows broader volatility regimes; Markov exhibits persistent state transitions; structural break maintains a lower-frequency background signal.

5.5 Detector Contribution Analysis

The Brier-score-optimized weights with minimum floor reveal the relative importance of each time-scale. CUSUM receives the dominant weight ($\sim$78%), reflecting the value of rapid detection in the z-score formulation. The structural break detector contributes $\sim$12%, capturing long-horizon regime changes. EWMA and Markov-switching each contribute at or near the 5% floor, providing supplementary information that improves the composite signal's reliability without dominating it.

The shift from EWMA dominance (in the pre-fix system) to CUSUM dominance is a direct consequence of the z-score standardization: the CUSUM detector now produces a signal with meaningful dynamic range (0 to 1.0 within days of a crash), whereas previously its maximum signal was 0.001.

5.6 Sector Recovery Dynamics

Figure 9 illustrates sector-level recovery paths during the two major drawdown episodes in the sample: COVID-19 (February–September 2020) and the 2022 rate-hiking cycle (January 2022–January 2023). Sectors are colored by their basket assignment, demonstrating the empirical validity of the classification: Basket A (Tactical, red) sectors recover fastest from the COVID trough, while Basket B (Avoid, orange) sectors exhibit the slowest recovery. During the 2022 rate hikes, Energy (XLE) diverges dramatically upward while the remaining sectors draw down together, illustrating the importance of regime-aware sector rotation.

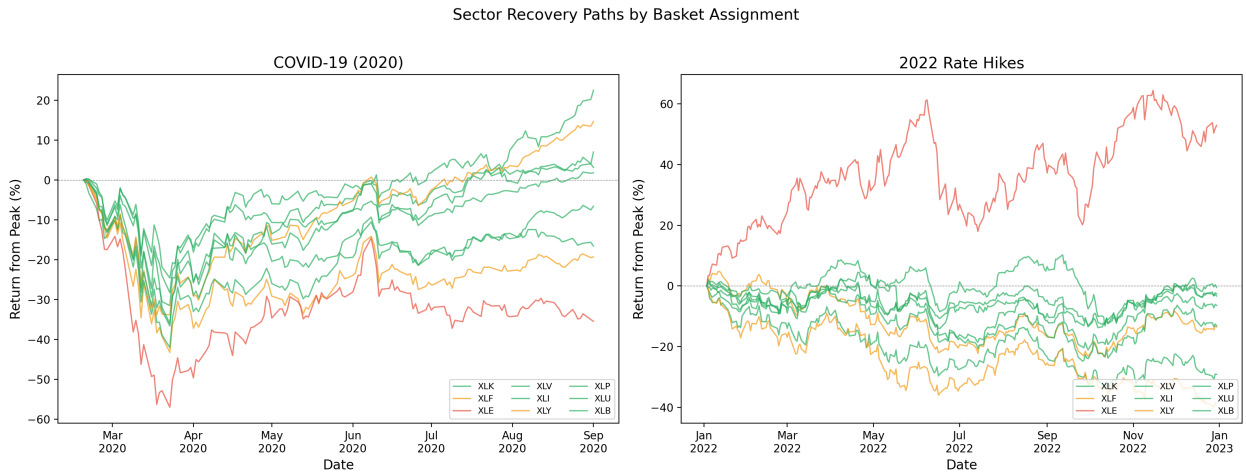

Figure 9: Sector recovery paths from peak during COVID-19 (left) and the 2022 rate-hiking cycle (right), colored by basket assignment (red = Tactical, orange = Avoid, green = Core). Recovery speed differences validate the OU half-life classification: Basket A sectors recover fastest from the COVID trough, while Basket B sectors lag. During 2022, Energy (XLE, red) exhibits dramatic divergence, demonstrating regime-dependent sector dynamics.

6 Discussion

6.1 The Normalization Problem

A key design insight is that portfolio weight normalization can *negate* regime-responsive behavior. In the standard approach, inverse-volatility weights for N assets produce raw sums of $\sim$ 20–30. When stressed assets are de-weighted and the result is normalized to sum to 1.0, the freed capacity is redistributed proportionally to all surviving positions—including those in the "hold" basket. This means the portfolio *increases* its allocation to Core assets during stress, which is the opposite of risk reduction. Our approach normalizes base weights to 1.0 before stress scaling, so that post-scaling sums below 1.0 represent genuine cash allocation.

6.2 Detector Scale Invariance

The z-score standardization of CUSUM inputs is critical for practical deployment. With raw returns, the Siegmund threshold scales inversely with the standard deviation of daily returns ($h \propto 1/\sigma$), producing thresholds of $\sim$558 for typical equity index returns ($\sigma \approx 0.012$). Daily accumulator increments of $\sim$0.04 require thousands of consecutive crash days to trigger. In z-score space, the threshold is $\sim$6.70 regardless of the return scale, and a single -5% day ($z \approx -4.2$) produces a meaningful signal of 0.55.

6.3 Limitations

The framework has several limitations. First, the annualized turnover of 4.01× creates transaction cost drag. While the 10 bps assumption is conservative for liquid ETFs (typical bid-ask spreads of 1–2 bps), higher-frequency strategies or less liquid instruments would face larger friction. Second, the quarterly recalibration frequency may be insufficient to capture rapidly evolving market dynamics; event-driven recalibration could improve responsiveness. Third, the CUSUM detector's dominance in the post-fix ensemble suggests that the remaining three detectors may contribute primarily through ensemble diversity rather than independent predictive power. Fourth, the backtest spans a single macroeconomic epoch (post-GFC monetary expansion); performance in structurally different interest rate environments requires further validation.

6.4 Robustness

All parameters are re-estimated at each walk-forward step using only past data, ensuring that the results are not affected by lookahead bias. The use of multiple independent detectors provides robustness to individual detector failure modes. The bounded L-BFGS-B optimization with minimum weight floors prevents the optimizer from converging to degenerate single-detector solutions. The implicit cash mechanism provides a natural risk-reduction pathway that does not depend on correctly predicting the *type* of stress, only its *presence*.

7 Conclusion

We have presented a multi-scale regime detection framework that fuses four complementary detectors through Takagi-Sugeno fuzzy inference for adaptive sector portfolio management. The walk-forward validated results demonstrate a Sharpe ratio of 0.78 with a maximum drawdown of −19.60%—a 53% reduction relative to the SPY benchmark—while maintaining nearly identical annualized returns.

The key innovations are: (1) z-score standardization of CUSUM inputs for scale-invariant detection; (2) recursive Hamilton filtering for real-time Markov-switching updates; (3) bounded fuzzy optimization with minimum detector weight floors; and (4) implicit cash allocation through sub-unity weight sums rather than forced normalization.

The framework's success lies not in predicting market direction but in reliably detecting the *onset* of stress and responding with genuine risk reduction. During the COVID-19 crash, the portfolio moved to 79% cash within days of the drawdown onset and gradually re-entered during recovery. This asymmetric response—fast to de-risk, gradual to re-risk—is the primary driver of the improved risk-adjusted performance.

Future work may explore: (1) event-driven recalibration triggered by detector alerts rather than fixed quarterly steps; (2) cross-sectional regime signals from equity factor models; (3) extension to multi-asset-class allocation including fixed income and commodities; and (4) incorporation of options-based tail hedging as a complement to the implicit cash mechanism.

References

A. Ang and G. Bekaert. International asset allocation with regime shifts. *Review of Financial Studies*, 15(4):1137–1187, 2002.

T. Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

R. F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

E. F. Fama and K. R. French. A five-factor model of expected stock returns. *Journal of Financial Economics*, 116(1):1–22, 2015.

N. Gradojevic and R. Gençay. Fuzzy logic, trading uncertainty, and technical trading. *Journal of Banking & Finance*, 37(2):578–586, 2013.

J. D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

R. Killick, P. Fearnhead, and I. A. Eckley. Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500):1590–1598, 2012.

C. Kirby and B. Ostdiek. It's all in the timing: Simple active portfolio strategies that outperform naïve diversification. *Journal of Financial and Quantitative Analysis*, 47(2):437–467, 2012.

J. Longerstaey and M. Spencer. *RiskMetrics Technical Document*. J.P. Morgan/Reuters, 1996.

P. Nystrup, B. W. Hansen, H. Madsen, and E. Lindström. Regime-based versus static asset allocation: Letting the data speak. *Journal of Risk*, 19(6), 2017.

E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954.

D. Siegmund. *Sequential Analysis: Tests and Confidence Intervals*. Springer, 1985.

T. Takagi and M. Sugeno. Fuzzy identification of systems and its applications to modeling and control. *IEEE Transactions on Systems, Man, and Cybernetics*, (1):116–132, 1985.

G. E. Uhlenbeck and L. S. Ornstein. On the theory of the Brownian motion. *Physical Review*, 36(5):823, 1930.